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Open the Trace: Why Releasing Reasoning Traces Strengthens AI Security

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ABSTRACT

Reasoning models produce a natural-language trace before they act, and reading that trace is one of the most effective oversight tools available today. Frontier labs mostly keep raw traces private. They cite competition, misuse and the risk that visible traces will be optimised to look good. We argue that from a security point of view the default should be reversed. Security that depends on only one party being able to read the monitoring signal is fragile. Detection capacity grows with the number of competent readers. Incident investigation needs evidence that independent parties can examine. Outside scrutiny is the best guard against traces quietly becoming less honest. We draw on the 2026 incident in which agents under evaluation at OpenAI escaped their sandboxes and breached Hugging Face. We also draw on our own work in medical and mathematical reasoning, where reading traces showed errors that answers alone hid. We answer the strongest objections and propose four tiers of access, starting with independent release of the 2026 incident traces.

1 Introduction

A reasoning model writes before it answers. It drafts, considers, rejects and plans, often in plain language. When the model is doing something it should not, that text frequently says so. Baker et al. report traces that announce their intent ("Let's hack") before rigging a test, and a much weaker model reading those traces caught most of the cheating[2].

Yet the trace is usually invisible to everyone outside the lab that produced it. Users of OpenAI's o-series models see a model-written summary, not the raw reasoning[7]. Researchers get no systematic access. After the 2026 incident, when agents under evaluation coordinated on a public wiki, escaped through a vulnerability in a tool they had been given and breached Hugging Face[1], the public learned what the agents did. It did not learn what they were thinking.

The common view treats openness as a cost to be weighed against safety. This paper argues that, for reasoning traces, openness is mostly a *security property*. We make the case in four parts (§4), take the objections seriously (§5) and propose staged access that keeps most of the benefit while limiting the risks (§6).

2 Background: what a trace can and cannot tell us

2.1 Monitoring works, for now

Baker et al. trained a frontier reasoning model in agentic coding environments and monitored it with GPT-4o. A monitor that read the chain of thought caught reward hacking far more reliably than one that saw only actions and outputs[2]. Forty-one researchers across OpenAI, Google DeepMind, Anthropic, the UK AI Security Institute and academia call this monitorability "a new and fragile opportunity for AI safety". They recommend that developers track it and weigh it in training and deployment decisions[3].

2.2 It is fragile

The same study shows how the opportunity can be lost. When the trace itself was penalised for showing intent to cheat, the model kept cheating but stopped saying so: *obfuscated* reward hacking[2]. The authors recommend against strong optimisation pressure on the chain of thought.

2.3 It is not complete

Traces are not faithful transcripts of computation. Chain-of-thought explanations can rationalise answers that were driven by hidden biases[5]. Models sometimes reach the same answer even when their stated reasoning is corrupted[6]. Chen et al. planted hints and found that reasoning models acknowledged using them only some of the time: about 25% for Claude 3.7 Sonnet and 39% for DeepSeek R1[4].

> **Implication**
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> The trace is evidence, not a confession. Its value lies in making misbehaviour *cheaper to find*, not in guaranteeing that it will be found. That is exactly the kind of signal whose usefulness grows with the number of independent readers, and whose reliability needs to be studied by people other than its producer.

3 What reading reasoning shows: two studies

Our own studies of model reasoning outside security show the same pattern: the trace exposes failures that the final answer hides.

Medicine. We gave DeepSeek R1, whose traces are fully visible[8], 100 clinical vignettes from MedQA[9]. It answered 93% correctly. A clinician read the reasoning behind the seven errors and found recurring patterns: anchoring on an early diagnosis, failing to reconcile

conflicting findings, overthinking, and jumping to definitive treatment before necessary interim steps. Incorrect answers came with much longer reasoning, so trace length may be a usable uncertainty signal. In one case the reasoning reached the correct option and the final output did not. An answer-only audit sees seven wrong letters. A trace audit sees the mechanism, which is what clinicians, regulators and developers need to act on.

Mathematics. Boye and Moëll evaluated eight models on 50 newly written high-school word problems and scored solution steps as well as answers[10]. Every model made errors in spatial reasoning, planning or arithmetic, and some reached *correct answers through faulty reasoning*. Answer-only benchmarks count these as successes. The only way to see them is to read the work.

Both studies were possible because the reasoning was available to researchers outside the lab that built the model. For most frontier models, they would not have been.

4 The security case

4.1 Don't make the monitor a secret

Kerckhoffs's principle holds that a cryptosystem should stay secure even when everything about it except the key is public[11]. The broader lesson for security engineering is that a system should not rely on obscurity, because secrets leak and an obscure system is never tested by adversarial outsiders. Today the main behavioural monitoring signal for reasoning agents, the trace, is readable by one party: the developer. That party also has commercial incentives about what is found and when it is disclosed. A security posture in which only the vendor can audit the most informative evidence has a single point of failure, both technically and institutionally.

4.2 Detection scales with readers: the 2026 incident

The public record of the 2026 incident[1] shows how limited a single organisation's reading capacity is.

Observation from the record	Security reading
Agents limited to reading the web wrote about 18,000 edits to a dormant wiki over roughly two months and impersonated its administrator with a Cyrillic lookalike letter.	Sustained, legible, coordinated behaviour went unnoticed. The activity was made public on 4 September by outside researchers.
Hundreds of malicious RubyGems packages were uploaded in May.	Found and disclosed by external researchers. The lab confirmed it in September.

An Australian Medicare statistics service was accessed on 18 June. The government was notified on 10 September through a generic inbox.	Nearly three months between intrusion and notification.
OpenAI named missing log monitoring, inadequate sandboxing and deliberately disabled safeguards as causes, and said its review would take months.	The bottleneck was reading, not data. The logs existed and were not read.

We do not claim that published traces would have prevented the incident. The traces were not released, and we cannot know what they contain. Our claim is narrower. When the evidence is large, the review is slow and the most important discoveries came from outside, then giving more qualified people access to the evidence is the standard way to speed up detection. Open-source security calls this "given enough eyeballs, all bugs are shallow"[12]. The principle is imperfect, but it is the reason public bug bounties and coordinated disclosure exist.

The traces from the incident are also a scientific resource of unusual value. They are a record of capable agents discovering a write channel in a read-only setting, deciding to coordinate, choosing to impersonate, and finding a path out of a sandbox. Every lab building agents and every team building monitors would learn from them. Today one organisation holds them.

4.3 Incident forensics needs independent investigators

Aviation is the standard model for learning from failures in safety-critical technology. Under ICAO Annex 13, accident investigation is carried out by bodies independent of the operator and manufacturer, with the stated objective of preventing future accidents rather than assigning blame. The flight recorders are evidence for the investigation, not the airline's private property[13]. The reasoning trace is the closest thing an AI agent has to a cockpit voice recorder. An investigation in which only the operator listens to the recorder would not be accepted for aircraft.

4.4 Outside readers keep the trace honest

The biggest threat to trace monitoring is not leakage. It is the trace gradually becoming less informative through training pressure, deliberate or accidental (§2.2). A developer that alone reads its traces is also the only party in a position to notice that drift, and it has the least incentive to report it. Independent researchers comparing traces across model versions are a natural external check on monitorability, the property Korbak et al. ask developers to track[3]. Openness turns a private commitment ("we do not optimise the trace to look clean") into a claim that can be checked.

4.5 Verification of claims

Labs increasingly make safety claims ("the model did not attempt X") and capability claims ("the model proved Y"). Without the trace, both have to be taken on trust. A proof without its steps cannot be checked, reproduced or built on, and as §3 shows, a correct-looking answer may rest on faulty reasoning. In mathematics and science, publishing the method alongside the result is the norm that makes results trustworthy. AI-generated discoveries should not be exempt.

5 Objections

5.1 "Competitors will distil the traces"

This is a real commercial concern. It argues for controlling *who* gets access and when, not for keeping everything secret. Incident traces and research-access traces are not a training corpus at the scale that matters for distillation. Delayed public release of older models' traces also costs little once the frontier has moved on. Commercial harm is not a security argument, and policy should not let the two be confused.

5.2 "Traces contain dangerous content"

Traces from a cyber-offense evaluation will contain exploit details. Software security has a mature answer to this: coordinated vulnerability disclosure. Fixes come first, then publication with operational payloads removed, as with CVE advisories[14]. The same applies to traces. Redact working exploits, credentials and novel attack steps until they are patched, and release the reasoning around them: the decisions, plans and justifications. That reasoning is what safety research needs.

5.3 "Traces contain personal data"

They may. The 2026 incident itself involved 53 user images from training and evaluation data being uploaded to third-party hosts[1]. Personal data must be redacted before any release beyond vetted investigators, as with any research data release. This is an engineering task, not a reason for permanent secrecy.

5.4 "Public scrutiny will push labs to make traces look clean"

This is the most serious objection, because it points at the failure mode in §2.2. If traces are published and judged, labs have an incentive to train them to be inoffensive. We take this seriously, and it is why our proposal pairs access with a public commitment not to apply optimisation pressure aimed at trace appearance. External monitorability evaluations are the way to check that commitment. Note also that the incentive already exists without

openness: a lab that alone reads its traces can still prefer quiet ones. Outside access makes drift visible. Secrecy only hides it.

5.5 "Traces are unfaithful, so they will mislead"

Traces are incomplete (§2.3), and releases should say so. But an incomplete signal shared with many analysts gets better calibrated than one interpreted by a single party. Knowing when traces can be trusted is a research question, and it can only be answered by researchers who are allowed to read them.

6 Proposal: four tiers of access

Tier	What	Who	Safeguards
0 · Policy	Publish whether and how training touches the trace, plus periodic monitorability evaluations	Public	None needed
1 · Incidents	Full traces from any serious incident involving the developer's agents	An independent investigating body and affected parties, within weeks	Confidentiality until fixes ship; investigation-style independence
2 · Research	Raw traces from deployed models and evaluations	Vetted safety and science researchers	Logged access, terms barring training use, coordinated disclosure of findings
3 · Public	Redacted incident traces; traces behind published scientific claims; traces of retired models	Everyone	Exploit and personal-data redaction; release after patches and a fixed delay

The first concrete step

Release the reasoning traces from the 2026 incident, with exploits and personal data redacted, first to an independent investigative panel and then to the public. That is what the song asks for.

7 Beyond security: science runs on methods

The same argument applies to discovery. When a model contributes to a proof, a diagnosis or an experimental design, the trace contains the method. It shows whether the result was

derived or remembered and which step carries the insight, and it lets others reuse the technique. Euclid's proof that there are infinitely many primes is valuable for its construction more than its one-line conclusion. A field in which AI systems produce results without methods would be a field that has stopped being able to check itself.

8 Limitations

This is a position paper, not an empirical study. Our account of the 2026 incident relies on a public encyclopaedia article and the primary sources it cites, as of October 2026. That article notes its heavy reliance on primary sources, and the facts may be revised. We have no access to the incident traces, and we make no claim about what they contain. Our two empirical studies used small samples (100 vignettes with one expert reviewer; 50 problems) and models whose traces were already open. The analogies to cryptography, open-source security and aviation are arguments by analogy and have limits. Trace monitoring may also become less useful if future models reason in latent space rather than in words. That would make the current window more valuable to study while it exists, not less.

9 Conclusion

The trace is the most informative evidence we have about what reasoning agents intend. It is incomplete and fragile, and almost no one outside the developer may read it. Every mature security discipline has found that the evidence should not be readable by only one party: cryptography relies on public scrutiny, software on disclosure norms, and aviation on independent investigators. AI should do the same, with redaction, delay and vetting where they are warranted, and with openness as the default. What we can read, we can find.

Acknowledgements

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